NORTHEASTERN UNIVERSITY
IE7615 Deep Learning

Beyond the Grid: A Trustworthy, Explainable Detector for 73 Everyday Objects



Discriminative Project: Milestone 3 Report

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Abstract

Milestone 2 delivered a YOLO26-s detector that reports the Object ID and bounding box of every object in grids of course photos, reaching mAP@0.5 = 0.947 on human-verified test boxes. Milestone 3 asked what it would take for that detector to be trustworthy beyond the conditions it was trained on. A diagnosis on validation data showed that turning a grid by 90° or 270° cut recall from 0.961 to about 0.77 and raised wrong Object IDs fourteen-fold (5 to 70), and the Milestone 2 model later found only 45% of objects placed on new backgrounds, evidence that it relied on the photos' surroundings. We first made the labels complete: in a purpose-built review tool, a person checked **every usable box in the dataset (6,881)** in two passes, corrected 423 boxes and removed 41 near-duplicate training and validation copies. We then built 3,000 training images with rotated and tilted photos, photos scattered on 385 real backgrounds and 6,752 objects cut out with SAM 2, and fine-tuned two models from the Milestone 2 detector, at 1,280 and 1,600 pixels. A selection rule fixed and hashed before the second model finished chose the 1,280-pixel model on validation (mAP@0.5:0.95 = 0.886 against 0.878; 95% bootstrap interval of the difference −0.014 to −0.002), and validated settings for grids and scenes (confidence thresholds 0.10 and 0.35, class-agnostic duplicate removal on scenes) raised recall further. On five test sets built from held-out photos with corrected truth, recall rose from 0.912 to **0.949** on normal grids, 0.756 to **0.922** on rotated grids, 0.767 to **0.897** on scattered scenes, 0.452 to **0.881** on cut-outs on unseen backgrounds and 0.905 to **0.951** on single photos, at the price of more wrong Object IDs on rotated grids. The interactive explorer, The EYE, was extended to every layout and model, with per-layout validated thresholds, drag-and-drop input and a box-tightness view; it is documented in a user manual and shared through a public link.

Keywords: object detection, YOLO, label verification, segmentation-based augmentation, rotation robustness, pre-registered model selection, explainable interfaces

At a Glance

6,881	0.45 → 0.88	0.76 → 0.92	0.91 → 0.95	1 rule
boxes, every one checked by a person	recall on objects placed on unseen backgrounds	recall on rotated grids	recall on normal grids	fixed in advance chose the final model

Table 1. Milestone 2 and Milestone 3 side by side.

	Milestone 2	**Milestone 3**
Boxes checked by a person	1,968 (every test box)	6,881 (every usable box), two passes
Near-duplicates across splits	not checked	183 pairs reviewed, 41 copies removed
Training images	1,500 upright grids and single photos	3,000: rotated grids, scattered scenes, SAM 2 cut-outs, empty backgrounds
Test sets	upright grids, single photos	normal, rotated, scattered, cut-outs, single photos
Detector	YOLO26-s, 1,120 px (run 5)	YOLO26-s fine-tuned from run 5, 1,280 px (run 6, boost)
Model selection	best validation mAP@0.5:0.95	a rule written and hashed before the result existed
Post-processing	one threshold (0.10), one box per cell	grids 0.10; scenes 0.35 with class-agnostic duplicate removal
Test recall, normal / rotated / scattered / cut-outs / singles	0.912 / 0.756 / 0.767 / 0.452 / 0.905	0.949 / 0.922 / 0.897 / 0.881 / 0.951
Explorer	The EYE v1	The EYE v3: every model and layout, validated thresholds, drag and drop, box tightness
Sharing	local only	public link through a tunnel, plus a user manual

All test numbers are on the same corrected truth for both milestones' models (Section 4.3); the Milestone 2 model scores mAP@0.5 = 0.947 there, as in its own report.

1. Introduction

At the end of Milestone 2 the detector was accurate on what it had seen: upright 224 x 224 photos placed side by side in grids. Three questions remained open. Would it still work if the image were turned, or if the objects were scattered on a table rather than arranged in cells? Were the remaining automatically accepted training boxes, about 3% of which were estimated to be imperfect, holding it back? And could anyone outside our laptop see, question and trust what it does? Milestone 3 answers all three.

Throughout, we kept the rules that made Milestone 2's numbers trustworthy: every decision on validation data, every test set used once per model, every comparison on the same truth. We added one more, because the milestone was carried out under a one-day deadline for the model: **the submitted Milestone 2 model, results, box file and explorer were never modified.** Every new script and output lives in a separate folder, training starts from a copy of the selected weights, and four SHA-256 hashes of the protected files were re-checked after every phase. They matched every time, so the Milestone 2 system remained a guaranteed fallback.

Contributions. This milestone makes six contributions:

- A **complete human check of every usable box** in the dataset (6,881), with a review tool that queues each kind of risk separately, a single global box rule, near-duplicate removal across splits, and a second pass that uses the improved detector as a second opinion (Section 3).
- **Training data beyond the grid**: rotated and tilted photos, photos scattered on real backgrounds, and objects cut out with SAM 2 and placed at any angle, with a geometry check of every label (Section 4).
- Two improved detectors and a **selection rule fixed and hashed before the result existed**, which chose the lower-resolution model on validation by a margin that a paired bootstrap shows is not noise (Sections 5 and 6).
- **Separate validated settings for grids and scenes**, including class-agnostic duplicate removal for scenes and a recall guard that only accepts a setting if it finds at least as many objects (Section 7).
- Five test sets on corrected truth, showing gains on every one, with the trade-offs stated (Section 8).
- **The EYE v3**, an explainable interface extended to every layout and model, a user manual, and a public link (Sections 9 and 10).

1.1 The model journey

Figures 1 and 2 map the whole project on two pages with the same four lanes: the models and data that made the labels, the human checks in our review tools, the detectors we trained and chose, and the versions of The EYE. Line styles say what each connection means: a detector starts from another's weights, trains on a set of labels or images, or gives a second opinion on labels. Read together, they show that labels, people and tools changed at every step alongside the models, and that every choice between models was made on validation data. Figure 3 condenses the Milestone 3 part into its eleven steps.

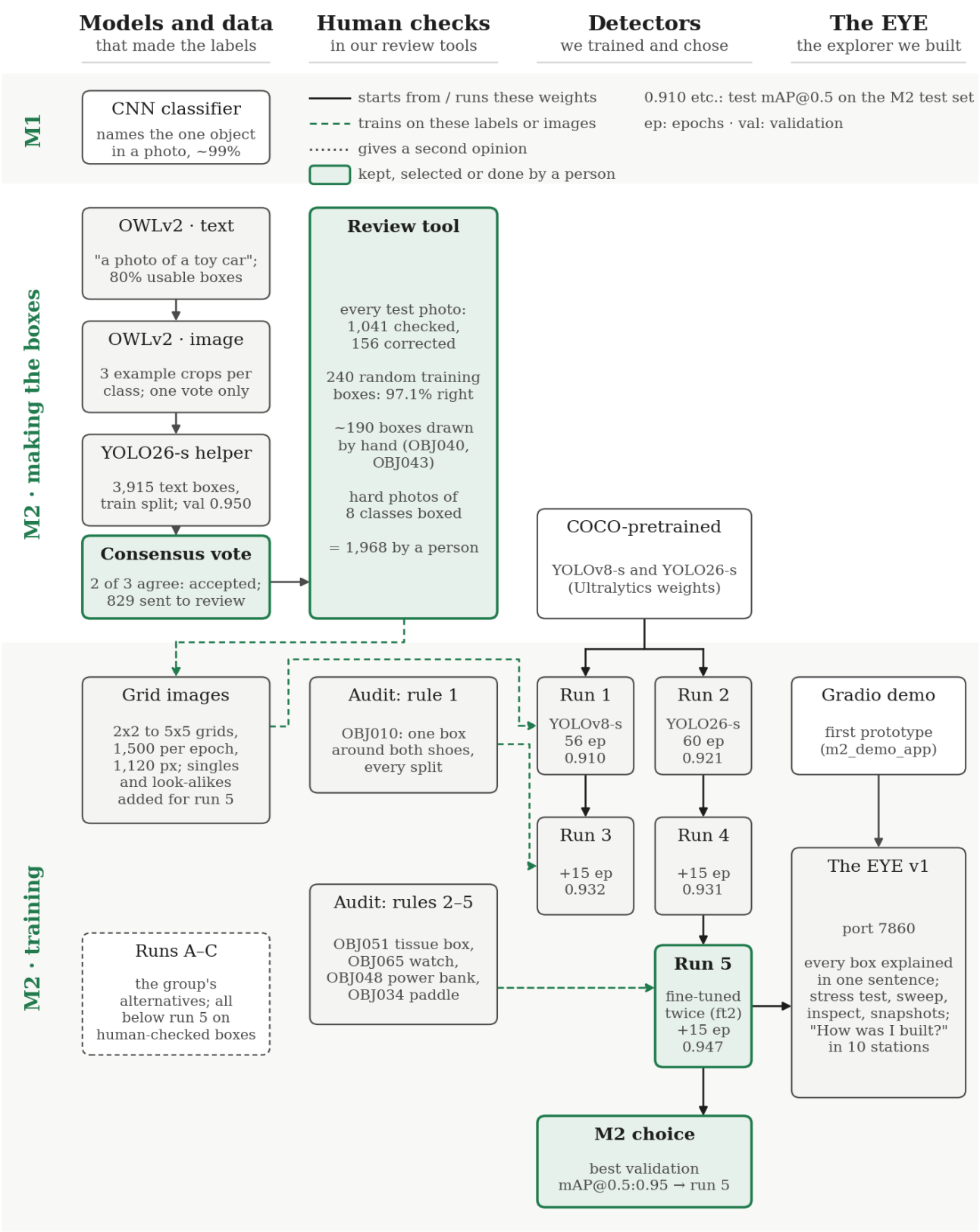


Figure 1. The model journey, part 1: Milestones 1 and 2, from the classifier to the Milestone 2 choice.

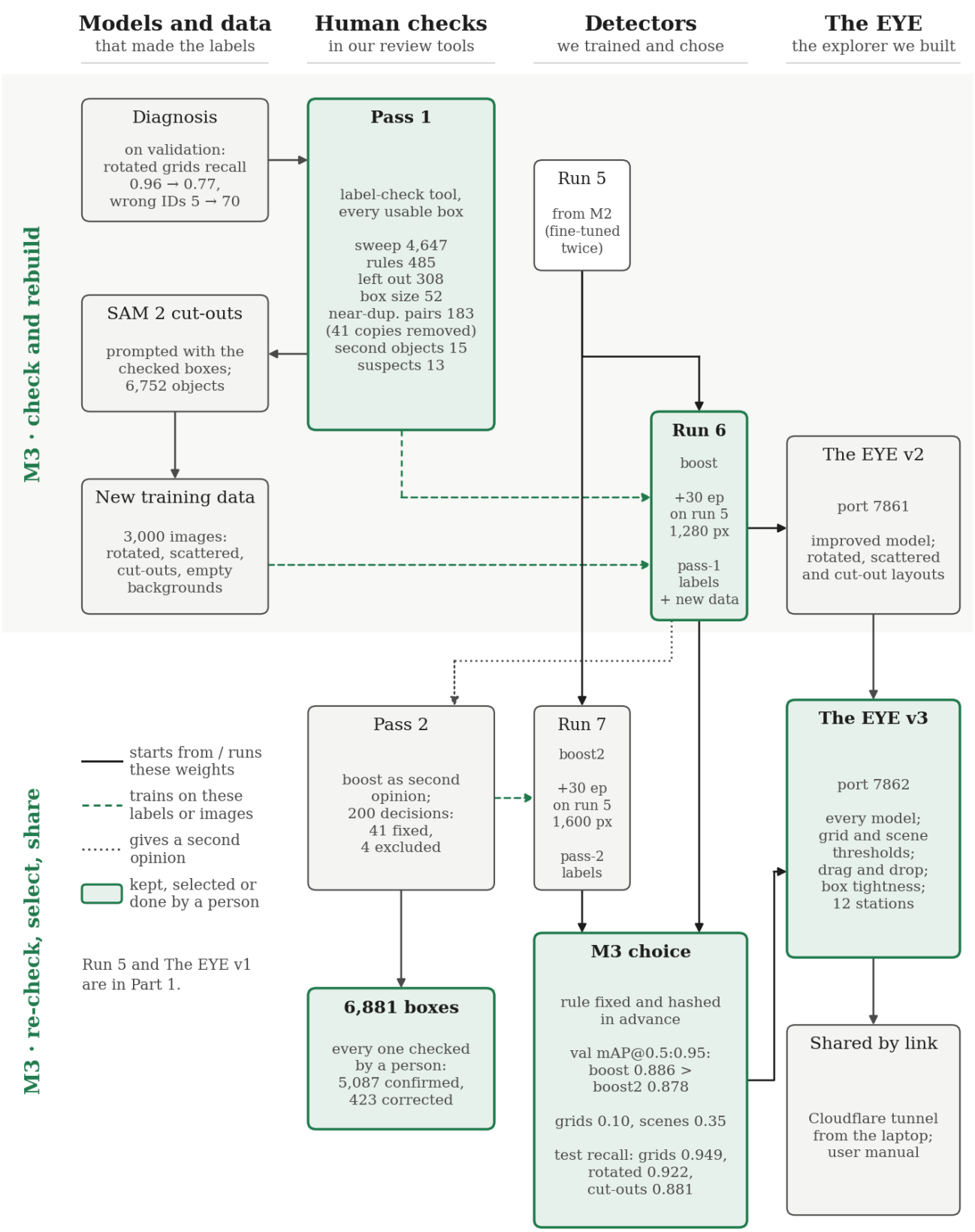


Figure 2. The model journey, part 2: Milestone 3, from the diagnosis to the shared explorer.

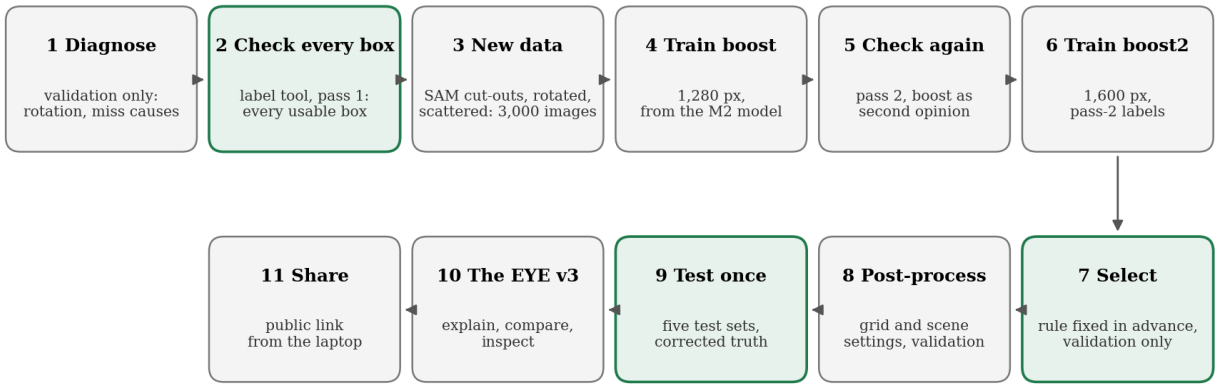


Figure 3. The Milestone 3 workflow. Highlighted steps carry the protocol: every box checked by a person, a model chosen by a rule fixed in advance, and one look at each test set.

2. Starting Point: What the Milestone 2 Model Could Not Do

Before changing anything, we measured where the Milestone 2 model (run 5, `yolo26s_ft2`) failed, using **validation data only** (68 grids, 972 objects), so that the diagnosis could not leak into the test results. On upright grids it found 96.1% of objects with 5 wrong Object IDs. Rotating the same grids exposed the main weakness (Figure 4, Table 2): recall fell to 0.771 at 90° and 0.762 at 270°, and wrong Object IDs rose from 5 to 70 (30 at 180°). The model had learned objects in their usual orientation; turned on their side, it named many of them as other objects.

Table 2. Diagnosis of the Milestone 2 model on validation grids.

Rotation	Recall	Wrong Object IDs	Comment
0° (upright)	0.961	5	precision 0.972, mAP@0.5 0.971
90°	0.771	70	objects on their side
180°	0.830	30	upside down
270°	0.762	70	objects on their other side

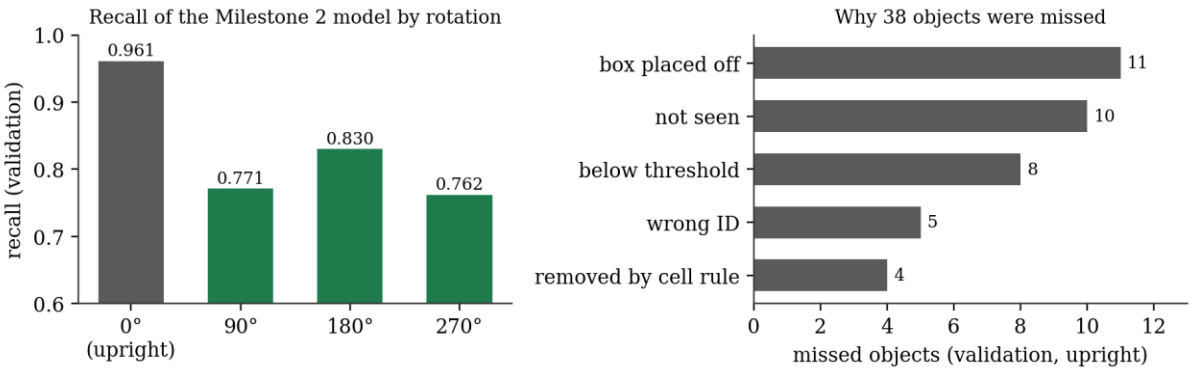


Figure 4. Left: recall of the Milestone 2 model by rotation (validation). Right: causes of the 38 missed objects on upright validation grids.

The 38 misses on upright grids had five causes. The largest, **box placed off** (11), pointed at labels rather than the model: partial or inconsistently drawn boxes. Ten objects were not seen at all (five of them very small), eight were seen below the threshold, five were given a wrong ID and four were removed by the one-box-per-cell rule. Three levers that need no retraining were also tested on validation and rejected: filling empty cells added 0.4 points of recall but cost 0.7 points of precision and two wrong IDs; test-time augmentation lowered the share of fully correct grids from 62% to 54%; and running the existing model at a larger input size without retraining was worse, as expected.

These findings set the plan: fix the labels first, then teach rotation and new layouts with new training data, and only then consider resolution. Because very small objects explained only about 5 of 38 misses, we trained the first improved model at 1,280 pixels rather than 1,600, which would have doubled the training time.

3. Every Box Checked by a Person

Milestone 2 checked every test box by hand but accepted 4,660 training and validation boxes automatically, after measuring that 97.1% of a random sample were correct. Milestone 3 removed that remaining uncertainty: **a person checked every usable box in the dataset.**

3.1 The label-check tool

We extended the Milestone 2 review tool into a label-check tool (`m2_label_check.py`) that sorts photos into queues, one per kind of risk, so that each kind of error is looked for where it is most likely (Figure 5). Every decision is written to disk immediately, so the review could be split across sessions, undone (with the saved box shown) and merged with teammates' decision files.

Table 3. Review queues of the first pass.

Queue	Photos	What is looked for
Sweep	4,647	every automatically accepted box, in a fast verification grid
Rules	485	photos of classes with a written box rule (sneakers, tissue box, watches, power bank, paddle)
Left out	308	photos left out in Milestone 2 because no automatic box was reliable; boxed by hand where possible
Near-duplicates	183 pairs	visually near-identical photos in different splits (87 test/train, 76 validation/train, 20 test/validation)
Box size	52 (24 test)	boxes much smaller or larger than the class's usual box: partial or oversized labels
Second objects	15	a second course object visible in the photo
Suspects	13	the self-trained model confidently sees a different object

A single global rule was written for every class: **box the whole object, every visible part, tight to its edges.** It generalises the five class-specific rules of Milestone 2 (for example one box around both shoes, or the whole watch with its strap) and removes the ambiguity behind the "box placed off" misses. The box-size queue found the same classes that the Milestone 2 audit had found, which confirms that audit: sneakers (10 cases), tissue box (5), table-tennis paddle (4) and analog watch (3).

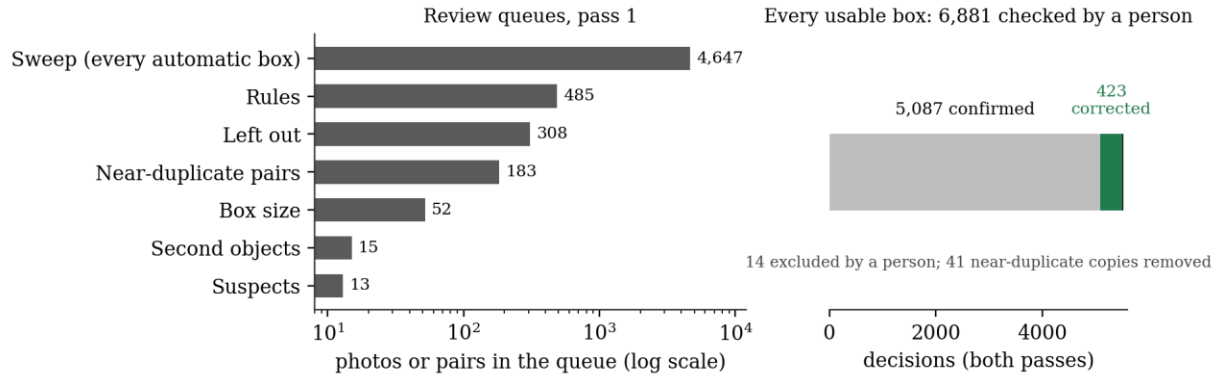


Figure 5. Left: size of each review queue in the first pass (log scale). Right: all decisions of both passes.

3.2 Near-duplicates across splits

Milestone 2 removed exact duplicates before splitting, but students sometimes took two photos a moment apart, which are not identical files but show the same scene. We compared every photo with every photo of the other splits by perceptual hash and reviewed the 183 closest pairs side by side. Where a test photo had a near-twin in training or validation, **the test photo was kept and its twin removed**, so the test set stays the same size and no test result can be inflated by a memorised scene; 41 training and validation copies were removed this way.

3.3 A second pass with the improved model as second opinion

After the first improved model was trained (Section 5), it was used as an independent second opinion on the corrected labels: wherever it disagreed with a box, or saw a different object, the photo went back to a person. This second pass took 200 decisions: 125 boxes confirmed, **41 corrected** (17 in the test set, 12 in validation, 12 in training), 4 photos excluded, and 30 second-object cases (11 boxes added, 19 dismissed). The tool replays all decisions of both passes in time order, so the final box file is reproducible from the decision log alone; a check against a backup of the first pass confirmed that no earlier correction was undone.

3.4 Outcome

Table 4. Labels before and after Milestone 3.

Measure	Milestone 2	Milestone 3
Photos with a box	6,628	6,881
Boxes checked by a person	1,968 (every test box and the review cases)	6,881 (every usable box)
Automatically accepted, unchecked	4,660	0
Boxes corrected by hand in this milestone	–	423
Photos excluded by a person	6 (not photos of the object)	14 more
Near-duplicate copies across splits	not checked	183 pairs reviewed, 41 copies removed
Second course objects labelled	0	16 boxes

Photos that had been left out in Milestone 2 were boxed by hand where the object could be seen, which raised the number of photos with a box from 6,628 to 6,881 even after the exclusions and the near-duplicate removal. Every box the models now train or are tested on has been seen by a person.

4. Data Beyond the Grid

4.1 Objects cut out with SAM 2

To teach the detector that an object is the same object on any background, we cut every object out of its photo. SAM 2 [18], prompted with the human-checked box, produced a mask for each photo; a quality gate compared the mask with the box (how much of the box the mask fills, how much of the mask spills outside it, and how well the mask's own box fits) and rejected 133 of 6,885 masks (1.9%). A visual check sheet confirmed that the masks followed the box rules: the tissue box with its tissue, the paddle without the hand, the watch with its strap. **6,752 cut-outs** were accepted.

4.2 Training images

The new training set has 3,000 images per epoch in four kinds (Figure 6), built only from training photos and training or synthetic backgrounds:

- **Grids (1,264)**, including about 15% single photos: individual photos turned by 90°, 180° or 270° or tilted by up to ±10°, whole grids rotated, and half of the grids with a complete look-alike group side by side, as in Milestone 2.
- **Scattered scenes (990)**: photos at random sizes and angles on real backgrounds, taken from the 385 background-only photos removed from the dataset in Milestone 1, or on synthetic backgrounds.
- **Cut-out scenes (597)**: cut-out objects at any angle and size, allowed to overlap partially.
- **Empty backgrounds (149)**, with no object, so that the detector learns not to box the background.

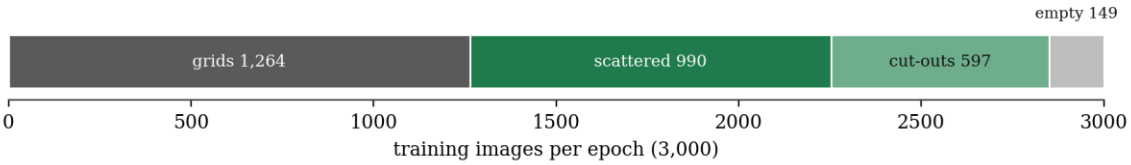


Figure 6. Composition of the 3,000 training images per epoch.

Photometric augmentation covers low light, colour casts, blur, JPEG compression and noise, the conditions Milestone 1 and 2 identified as hardest. Look-alike groups and the ten weakest validation classes are oversampled, and second course objects are labelled where they appear. Two placement rules keep every label honest: a later photo may cover at most 10% of an earlier object, and a later cut-out at most 20%, **cumulatively**, so no label ever describes an object that is mostly hidden. The rule was added after testing found that the first version could hide earlier objects.

Every label was verified geometrically: each object was rendered in a unique colour and its label compared with the largest connected region of that colour. The median IoU between label and object was 0.968 for grids, 0.967 for scattered scenes and 0.998 for cut-outs. Images are composed at the 1,120-pixel scale of the largest grid and scaled to the model's input size, in the same way for training, validation and inference.

4.3 Validation and test sets

Validation and test sets follow the Milestone 2 split exactly: validation scenes use only validation photos and backgrounds, test scenes only test photos and **backgrounds never used in training**. The test truth uses the corrected boxes, second objects are labelled, and cells of photos excluded by a person are ignored rather than counted (Table 5).

Table 5. Validation and test sets of Milestone 3.

Set	Images	Objects	Built from
Validation	292	3,465	136 grids (upright and rotated) and 156 scattered or cut-out scenes
Test: normal grids	72	1,048	the Milestone 2 test grids, with corrected truth
Test: rotated grids	216	3,144	each test grid at 90°, 180° and 270°
Test: scattered scenes	104	1,039	test photos on unseen backgrounds
Test: cut-outs	104	1,014	test cut-outs on unseen backgrounds
Test: single photos	1,040	1,048	every test photo on its own

Because the test truth was corrected, the numbers in this report are not directly comparable with the 0.947 of the Milestone 2 report: the Milestone 2 model itself scores mAP@0.5 = 0.947 on the corrected normal grids, but its recall and error counts change slightly. Every before-and-after comparison below therefore scores both models on the **same corrected truth**.

5. Training Two Improved Models

5.1 Settings

Both improved models were fine-tuned from a copy of the Milestone 2 model, so that what it had learned was kept and only the new skills had to be added (Figures 1 and 2 show the whole lineage). Mosaic augmentation was switched off because the new images are already composites.

Table 6. Training settings of the two improved models.

Setting	boost (run 6)	boost2 (run 7)
Start	copy of yolo26s_ft2	copy of yolo26s_ft2
Labels	after pass 1	after pass 2 (41 more corrections)
Input size	1,280 px	1,600 px
Batch size	4 (6.5 of 8 GB)	2 (6.4 of 8 GB; Section 5.2)
Epochs	30 (flat from about 23)	30 (best: 28)
Learning rate	0.002, cosine, 1 warm-up epoch	same
Augmentation	horizontal flip 0.5, light scale, translation and colour; no mosaic	same
Early stopping	10 epochs without improvement	same
Time per epoch	about 4 min	about 5.8 min

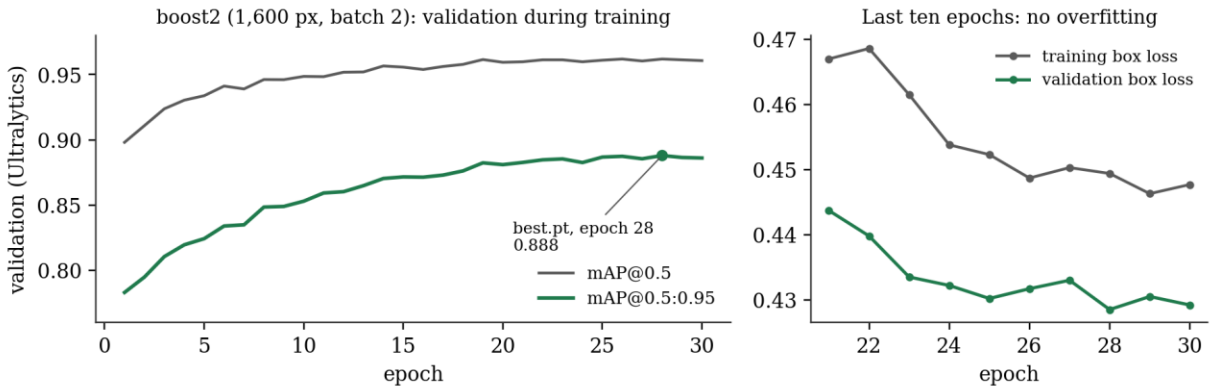


Figure 7. Training of boost2. Left: validation mAP during training (Ultralytics' own validation). Right: box loss in the last ten epochs: both fall, so the model did not overfit.

Both runs converged inside the 30 epochs. boost was flat from about epoch 23, with validation mAP@0.5:0.95 between 0.878 and its highest value of 0.880 at epoch 30; boost2 peaked at epoch 28, with mAP@0.5:0.95 between 0.886 and 0.888 over the last six epochs. Training and validation box loss both kept falling, so neither model overfitted. More epochs would have added at most a few thousandths, below the epoch-to-epoch noise.

5.2 A memory finding at 1,600 pixels

The first attempt to train boost2 used batch 4 and needed 9.42 GB on an 8 GB GPU. On Windows this does not fail: the driver silently moves the overflow into system memory over PCIe, and each training step took about 5.7 seconds instead of 0.25, a 22-fold slowdown for only 1.6 times more pixels, which projected to 32 hours. The run was stopped in its first epoch, before any result existed, and restarted at batch 2 (5.0 GB reported, 6.35 GB dedicated, 0.23 s per step). The restart is recorded in a note next to the selection rule. The episode also exposed a flaw in our automatic fallback to a smaller batch, which waited for an out-of-memory error that Windows never raises; a correct check compares reserved memory with the card's capacity.

6. Model Selection by a Rule Fixed in Advance

With two improved models, choosing the better one by looking at five test tables would have been a decision made on test data. While boost2 was still training, and before any of its results existed, we therefore wrote the selection rule into a file and recorded its SHA-256 hash: **boost2 replaces boost only if its mAP@0.5:0.95 on the boost2 validation set (292 images, pass-2 labels) is higher, each model at its own input size; thresholds and post-processing are chosen on validation only; test sets are reported for the selected model and do not influence the choice.** The script that applies it checks the hash before running.

The comparison was scored with exactly the same matching and AP code as the evaluation, imported rather than copied. To judge whether a difference is real, we added a paired bootstrap over validation images: 1,000 resamples, each scoring both models on the same resampled images. A fast, resamplable implementation of mAP was verified to reproduce the evaluation code exactly on resampled sets built by duplicating images.

Table 7. Model selection on the boost2 validation set.

Model	Input size	Batch	mAP@0.5	mAP@0.5:0.95
boost (selected)	1,280	4	0.960	0.886
boost2	1,600	2	0.952	0.878

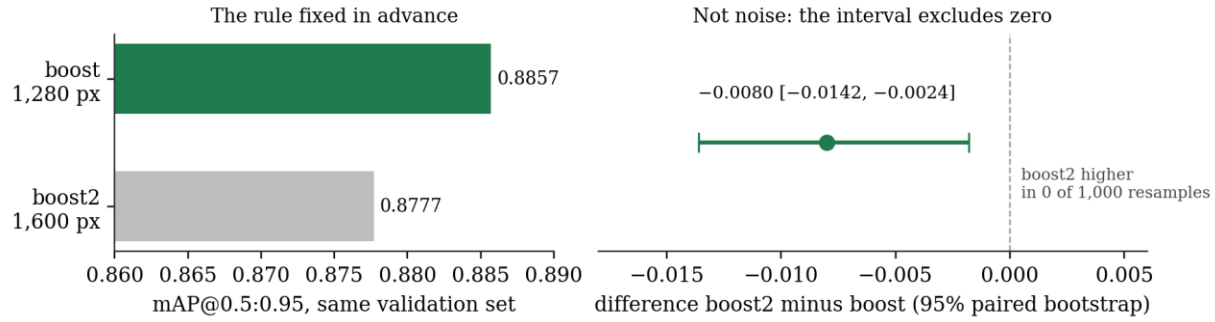


Figure 8. Left: the quantity named by the rule. Right: the difference with its 95% paired bootstrap interval.

The rule kept boost. The difference, −0.0080, has a 95% interval of −0.0142 to −0.0024, and boost2 was higher in none of the 1,000 resamples, so the result is not noise. This reverses the impression given during training, when Ultralytics' validation showed boost2 at 0.888 against boost's best of 0.880: those two numbers came from different validation sets (pass-1 and pass-2 labels), whereas the rule scores both models on the same set with the same code. We report this as a finding rather than a disappointment: **raising the input size from 1,280 to 1,600 pixels gave no measurable gain on this dataset.** The likely reasons are that both models are close to the precision limit of the labels themselves (Section 8.4), that small objects caused few misses, and that batch 2 makes batch-normalisation statistics noisier; the comparison also changes labels, resolution and batch size together, so it is not a clean ablation of resolution.

7. Post-processing for Grids and Scenes

A confidence threshold balances two errors: a low threshold lets weak extra boxes through, a high one drops objects the model saw with moderate confidence. Milestone 2 chose one threshold for grids. With scenes added, one threshold became a compromise, because the two kinds of image are protected differently. On grids, the one-box-per-cell rule already removes most extra boxes, so a high threshold mainly removes correct but uncertain boxes. Scenes have no cells, so the threshold is their only defence, and YOLO's own duplicate removal works per class: when the model is unsure which of two watches it sees, it can keep both boxes on the same object, one of them a wrong ID.

We therefore let validation choose, separately, a threshold for grids, a threshold for scenes and, for scenes only, **class-agnostic duplicate removal** (where boxes of any class overlap above an IoU of 0.5, 0.6 or 0.7, keep only the most confident; or none), the scene counterpart of the cell rule. Two true objects in our scenes overlap by at most 20% of one footprint (IoU about 0.11), so this rule cannot merge real neighbours. The search covered thresholds from 0.05 to 0.60, extending the evaluation's list, whose top value of 0.40 was exactly where the single threshold had landed. A setting was adopted only if, on validation, **F1 rose by more than 0.003 and recall did not fall** below that of the single threshold; the recall guard was part of the search, and both conditions were fixed, and the script's hash recorded, before any boost2 result existed.

Table 8. Post-processing chosen on validation (selected model, 136 grids and 156 scenes).

Setting	Thresholds	Recall	Precision	F1	Wrong IDs
One threshold (as in the evaluation)	0.40	0.929	0.963	0.946	39
Grids and scenes, scene duplicate removal at IoU 0.5 (adopted)	0.10 and 0.35	0.937	0.965	0.951	46

The grid threshold fell to 0.10, the value Milestone 2 had also chosen for grids, and scenes settled at 0.35 with duplicate removal. On validation this found more objects and kept precision, at the cost of seven more wrong IDs. The same settings are used in the evaluation and in The EYE, which switches between them automatically for the image on screen; we verified that The EYE's implementation of the scene rule gives identical results to the evaluation's on 300 random scenes.

8. Results

8.1 Every test set, before and after

Table 9 compares the Milestone 2 model with the selected model and its final settings on all five test sets, each used once, on the same corrected truth. mAP does not depend on the threshold and is reported from the raw detections.

Table 9. Test results: the Milestone 2 model and the selected model with its final settings (bold), same corrected truth. mAP50 = mAP@0.5, mAP50-95 = mAP@0.5:0.95.

Test set	Model	Recall	Precision	Wrong IDs	Fully correct	Small-object recall	mAP50	mAP50-95
Normal grids	M2 model	0.912	0.950	16	0.347	0.750	0.947	0.873
	Final	**0.949**	**0.970**	**10**	**0.472**	**0.896**	**0.971**	**0.900**
Rotated grids	M2 model	0.756	0.845	230	0.093	0.458	0.813	0.714
	Final	**0.922**	**0.947**	**58**	**0.347**	**0.747**	**0.953**	**0.863**
Scattered	M2 model	0.767	0.830	33	0.086	0.411	0.803	0.635
	Final	**0.897**	**0.943**	**17**	**0.250**	**0.667**	**0.936**	**0.803**
Cut-outs	M2 model	0.452	0.799	70	0.000	0.157	0.559	0.527
	Final	**0.881**	**0.914**	**61**	**0.288**	**0.590**	**0.912**	**0.887**
Single photos	M2 model	0.905	0.954	12	0.904	0.698	0.941	0.865
	Final	**0.951**	**0.968**	**11**	**0.951**	**0.896**	**0.970**	**0.894**

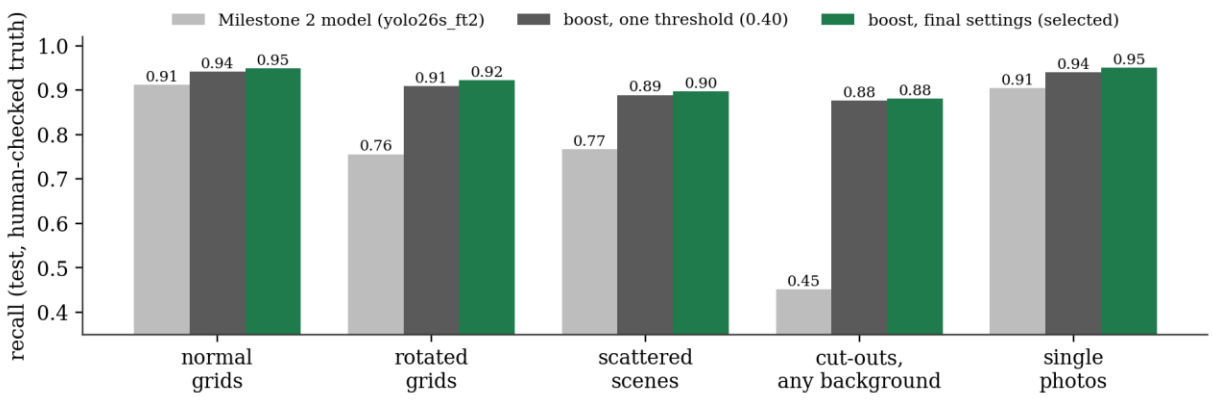


Figure 9. Recall on each test set: the Milestone 2 model, the selected model at the evaluation's single threshold, and the selected model with the final settings.

The improvement holds on **every test set and every metric**. Rotation, the main weakness of the diagnosis, is largely solved: recall on rotated grids rose by 16.6 points and wrong IDs fell from 230 to 58. The largest change is on cut-outs: the Milestone 2 model found 45% of objects placed on new backgrounds and the selected model 88%, which confirms that the earlier model had relied on each photo's surroundings. Normal grids and single photos, the conditions Milestone 2 was built for, improved as well, so the new skills did not cost the old ones; small objects gained most (normal grids 0.750 to 0.896).

8.2 What the final settings changed

Compared with the same model at the evaluation's single threshold of 0.40, the grid and scene settings raised recall on every test set, by 0.5 to 1.3 points (normal grids 0.942 to 0.949, rotated 0.909 to 0.922, scattered 0.889 to 0.897, cut-outs 0.876 to 0.881, single photos 0.940 to 0.951), and raised precision on cut-outs (0.898 to 0.914). The price is more wrong IDs, most visibly on rotated grids (38 to 58 of 3,144 objects): the lower grid threshold keeps more uncertain boxes, and some of them name the wrong look-alike. We accepted this because the task is to find every object, and state it rather than hide it.

8.3 The second improved model on the same tests

For completeness, boost2 was also evaluated on every test set (Appendix B). The two models are practically tied: boost2 has slightly higher recall on four of five sets (by at most 0.007) and slightly lower mAP@0.5:0.95 on four of five. The test tables would therefore not have given a clear answer, which is exactly why the choice was made on validation by a rule fixed beforehand.

8.4 Qualitative result and box tightness

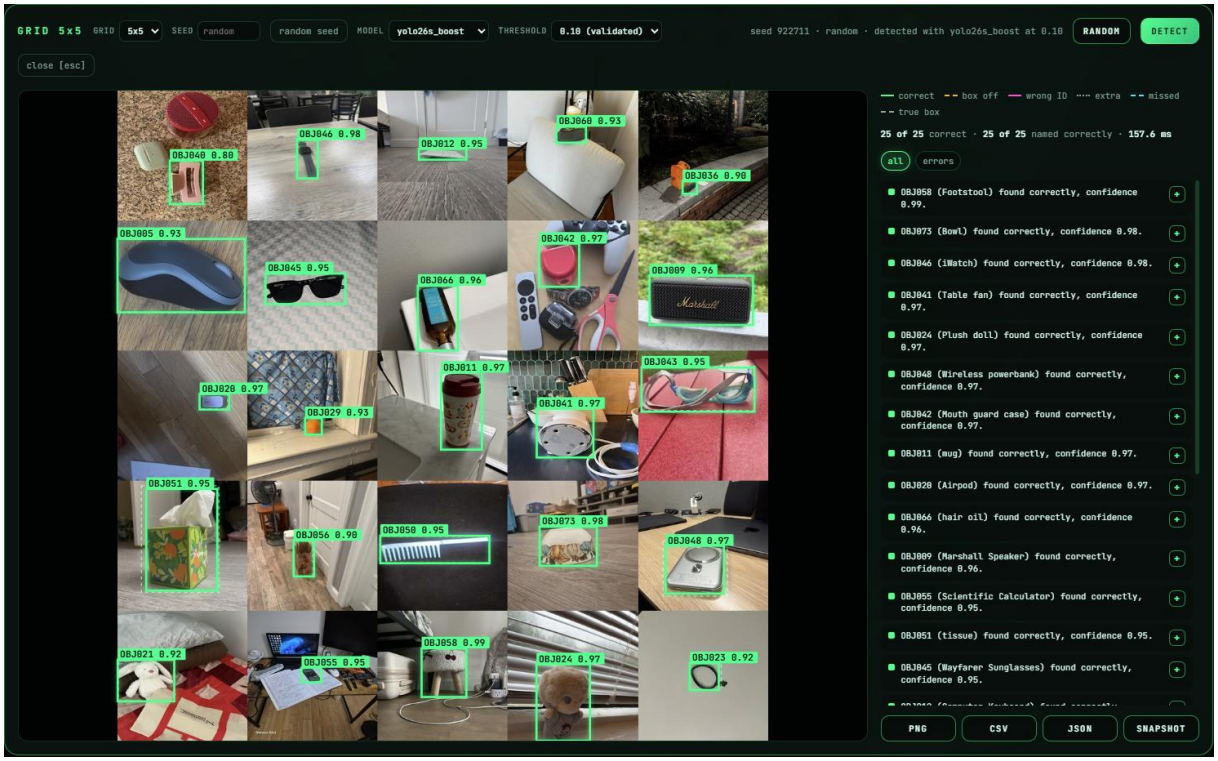


Figure 10. The selected model on a random 5x5 test grid (seed 922711) in The EYE's large view: all 25 objects found with the right Object ID and a correct box, in 157.6 ms. The list on the right gives one sentence per object; every row expands to its full record.

Why mAP@0.5:0.95 is lower than mAP@0.5. The stricter metric also asks for boxes that overlap the true box by 0.75, 0.90 and 0.95. On grids, most boxes are already that tight: in the box-tightness view of a 5x5 test grid (Figure 14, panel a), 17 of 25 boxes reach an IoU of 0.95 and the median is 0.96, and the final model reaches mAP@0.5:0.95 = 0.900 on normal grids. The credit lost at the strict thresholds comes from a minority of looser boxes, on small objects and above all in rotated and scattered images, where an axis-aligned box around a tilted object is loose by construction (scattered scenes: 0.803). The labels themselves match their objects at a median IoU of about 0.97 (Section 4.2), which sets a ceiling no detector can pass against them; mAP@0.5 and recall remain the measures of whether objects are found and named.

8.5 Robustness at scale

A test set is one sample; The EYE's stress test draws many. It builds random grids from held-out test photos with fixed seeds (1000, 1001, ...), so every run can be repeated exactly, and reports the share of objects correct, the share of fully correct grids with a bootstrap interval, and the time per grid. On 100 random 3x3 grids, the selected model at its validated threshold found **95.3% of objects correctly (858 of 900)** and named 97.4% correctly; **65.0% of grids were entirely correct** (95% CI 55.0 to 74.0%), at 42 ms per grid on the laptop GPU (Figure 13, panel a). This agrees with the 95% recall on the normal test grids. The run scores against the explorer's box file, which does not yet include this milestone's label corrections (Section 12.1), so it complements rather than replaces Table 9.

8.6 What still goes wrong

The same stress test also classifies every error. Of the 42 objects that were not fully correct in 100 random 3x3 grids, 19 had the right Object ID but a box placed off, 18 were missed and 5 got a wrong ID; 6 extra boxes appeared. The causes listed by the explorer (Figure 13, panel a) point to four groups:

Table 10. Error taxonomy from the stress test (100 × 3x3, final model).

Cause	Cases	What it means	Lever
Right ID, box placed off	19	the box covers part of the object or more than it	box definitions; the pass-2 corrections the explorer does not use yet
Very small object	8	the object fills a few percent of its photo	a higher input size did not help (Section 6); a P2 head might
Seen below the threshold	5	found, but with low confidence	a lower threshold, at the cost of extra boxes
Box on the background	4	often a second, unlabelled course object	label co-occurring objects

The objects most often missed or wrong were OBJ061 (card wallet, 5 times), a look-alike of the men's wallet, and OBJ026, OBJ059, OBJ063 and OBJ069 (3 each). Each confusion occurred only once, among them sunglasses for wayfarer sunglasses (OBJ014 → OBJ045), a known look-alike pair. The pattern matches the diagnosis of Section 2: **box placement, not naming, is now the largest source of error**, so further gains are more likely to come from labels and box definitions than from a larger model. And the share of fully correct images stays modest by construction: with about 95% of objects correct, a 16-object grid is entirely correct only about $0.95^{16} \approx 44\%$ of the time.

9. The EYE v3

The EYE (Explainable YOLO Explorer) was introduced in Milestone 2 as a local interface in which every result can be questioned. In this milestone it was extended in two steps, v2 and v3, into a tool that shows every model on every layout, applies exactly the settings of the evaluation and explains each result. The original EYE and v2 were kept unchanged as fallbacks.

9.1 What changed

Table 11. The EYE: what Milestone 3 added.

Area	Change
Models	every trained model in one list; the model chosen on validation is the default ("selected"), the Milestone 2 model is marked "M2 pick"
Layouts	Mix panel chips for grid, rotated (a live test grid turned 90°, 180° or 270°), scattered and cut-out test scenes
Thresholds	each model's validated values, switched automatically for the image on screen (grids 0.10, scenes 0.35, scene duplicate removal as in the evaluation); a hand-set value is kept until reset
Input	drag and drop any photo onto the page; non-images are refused
Box tightness	a second view in the sweep panel: IoU against confidence for every box, with the 0.50, 0.75 and 0.95 cut-offs
About and journey	before-and-after tables on every test set, the selection result, and twelve construction stations instead of ten
Sharing	a configurable cap on the stress test, so one visitor cannot block the shared GPU

Smaller fixes made the interface sturdier without changing what it shows: panels never hide their content on any screen size, buttons wrap instead of overflowing, and the DETECT button has its own slot on the score ring.

9.2 The EYE in use

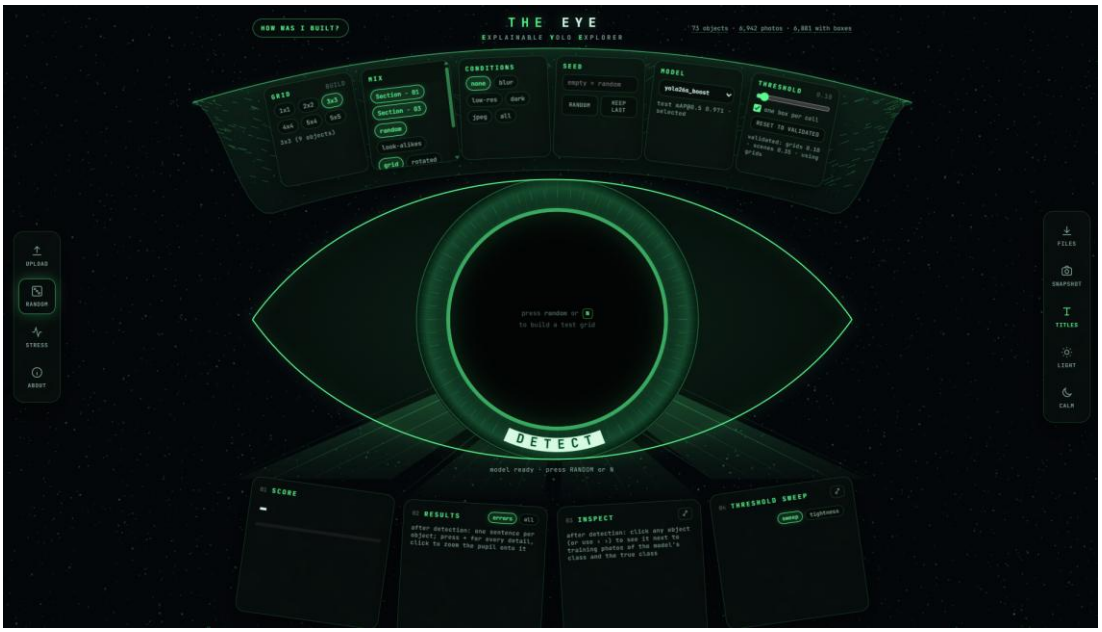


Figure 11. The EYE v3 before a detection. The brow holds the test set-up: grid size; the Mix panel with sections, random or look-alike objects and the layout chips; image conditions; seed; model (the selected yolo26s_boost); and the threshold, whose note shows the validated values for grids (0.10) and scenes (0.35). The counts at the top right include the 6,881 photos with checked boxes.

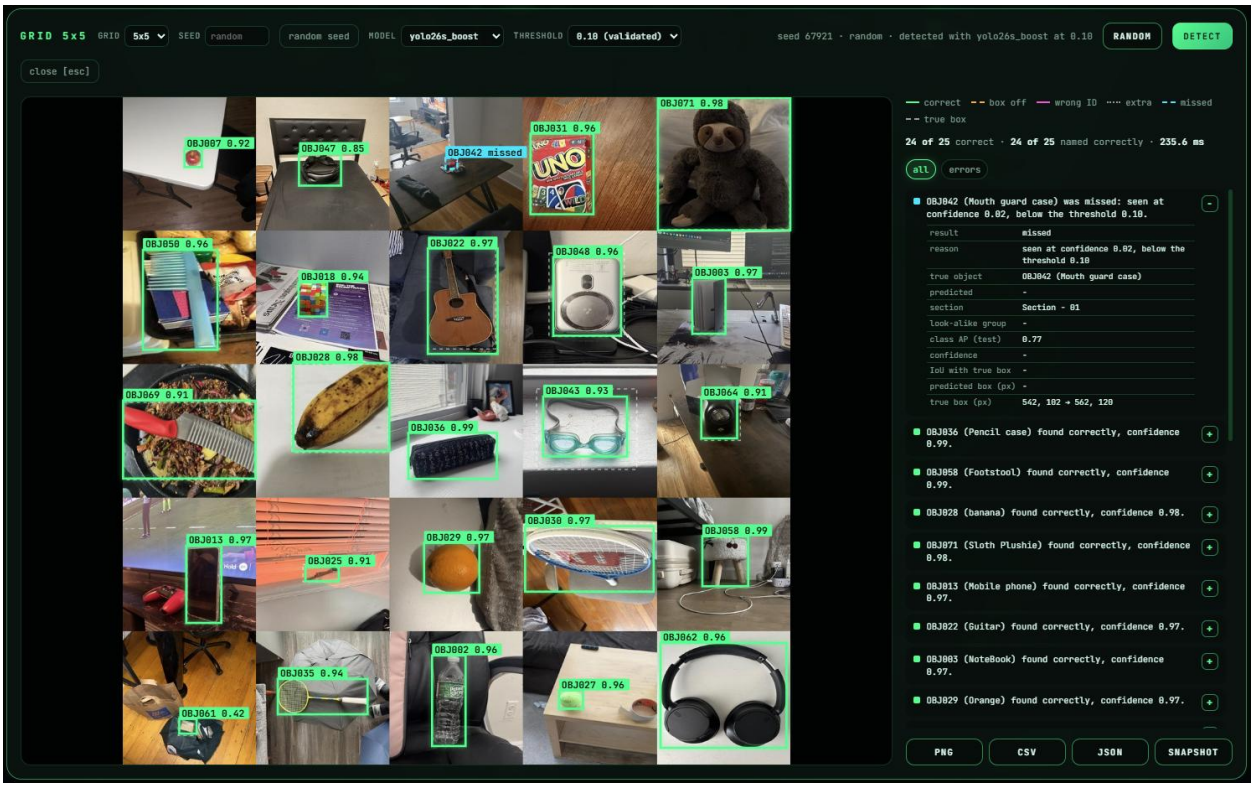
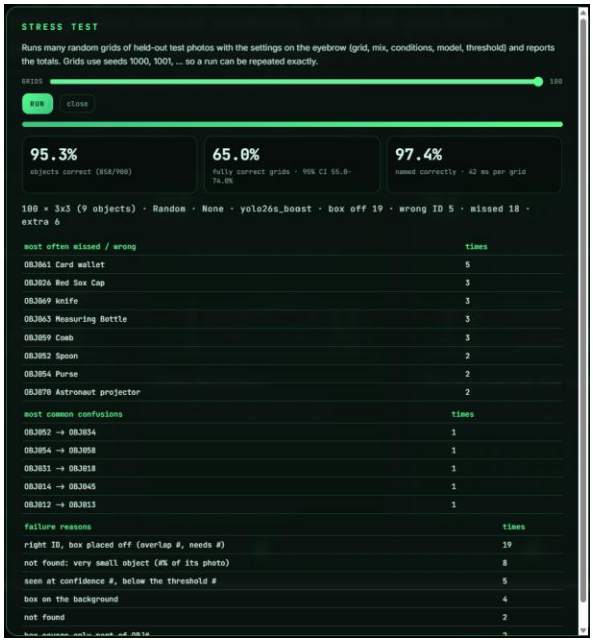
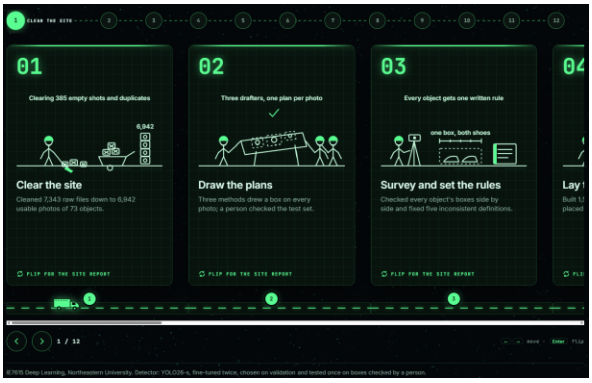


Figure 12. Large view after detecting a random 5x5 test grid: 24 of 25 objects correct. The one miss is explained in words: OBJ042 (mouth-guard case) was seen at confidence 0.02, below the threshold 0.10; its record lists section, class AP and the true box. Results download as PNG, CSV or JSON, or as a report snapshot.

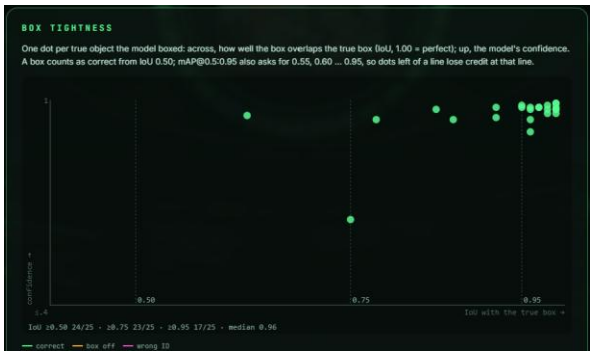


(a) Stress test: 100 random 3x3 grids with the selected model; 95.3% of objects correct, 65.0% of grids fully correct (95% CI 55.0–74.0%), 42 ms per grid, with the most missed objects, confusions and failure reasons.



(b) "How was I built?": the project as twelve stations on a construction site, one flip card per step, with the full site report on the back.

Figure 13. Two views beyond a single image: the stress test and the project story.



(a) Box tightness for a 5x5 test grid: 24 of 25 boxes reach IoU 0.50, 23 reach 0.75 and 17 reach 0.95; median IoU 0.96.



(b) Threshold sweep: correct objects, misses, extra boxes and wrong IDs at every threshold from 0.05 to 0.90; the dashed line marks the validated 0.10.

Figure 14. Two diagnostic views of one detection, both computed from the detections already made, without running the model again.

9.3 How it was tested

Every change was tested in a real browser against a synthetic project with stand-in models whose errors are known in advance, so the tests check what the screen must show rather than only that it runs: 150 automated checks across both themes and six screen sizes from 1,100 to 1,920 pixels wide, including a deliberately wider font. The explorer's scene rule was verified to give exactly the evaluation's results on 300 random scenes. The original EYE and v2 remained byte-identical.

9.4 User manual

A user manual for teammates and visitors documents every control, input and setting, the colour legend, how to read each panel, the stress test, the downloadable files, keyboard shortcuts, a glossary and troubleshooting; each behaviour in it was checked against the code. It is reproduced in Appendix C.

10. Sharing The EYE

10.1 How we shared it

The EYE runs on the laptop that holds the model and the photos, and a free Cloudflare quick tunnel [22] gives it a temporary public https address: no account, no firewall change and no change to the code. Two scripts start and stop sharing. The start script reuses a running EYE or starts one with a stress-test cap of 100 grids, opens the tunnel, retries with another protocol on networks that block the default one, checks the public address from outside and copies it to the clipboard; the stop script stops only the processes it started. Visitors get the full explorer at the speed of the laptop's GPU. The link exists only while we run it and changes every session, and it is shared only within the course.

10.2 Why not a permanent host

A permanent, always-on address was evaluated and deliberately not used, for three reasons:

- **Cost.** The EYE is a Python server that keeps each image, its detections and running stress tests in memory between requests, so it needs a host that runs one program continuously. Hugging Face, the

natural home for such a demo, now requires a paid plan for Docker Spaces [19], and GPU hardware there is billed by the hour; free CPU hosting would also make each detection several seconds slower.

- **Size.** PyTorch and Ultralytics alone take several hundred megabytes before the model weights and photos are added. That exceeds the 500 MB limit of serverless Python functions such as Vercel's [21], whose short-lived instances would also lose the server's state between requests; static hosts such as GitHub Pages cannot run the model at all. Free cloud sessions such as GitHub Codespaces [20] work but stop when idle, so they give no permanent link either.
- **Privacy.** The EYE shows the course's photographs, taken by students of their own belongings, often in their rooms. An always-on public address would publish them to anyone who finds it. A tunnel that exists only while we run it, shared within the course, keeps that exposure small, and wider publication would need the permission of everyone who contributed photos.

So that a permanent host can follow quickly once these conditions change, we prepared a self-contained bundle for a Docker Space: a script copies exactly the files The EYE reads, strips every local path, stores the test photos so that detection input is **pixel-identical** to the laptop (verified on 48 cases, including phone photos with rotation flags), caps the stress test and test-runs the bundled copy on every layout before upload.

11. Process Safeguards

Work under a short deadline is where mistakes enter, so we relied on procedures rather than on care alone:

- **A protected fallback.** The submitted model, results, box file and explorer were hashed before the milestone and re-checked after every phase; all four matched every time. New work wrote only to a separate folder, and training started from copies of the weights.
- **Decisions fixed before results.** The model-selection rule and the post-processing rule were written, and their files hashed, before the corresponding results existed; the selection script refuses to run if the rule file changed.
- **Resumable, logged steps.** Training resumes from its last checkpoint and refuses to overwrite a finished run; every review decision is logged and replayable.
- **Tests with known answers.** Every script was first run on a synthetic project with stand-in models whose errors are planted, and checked on its output files, not only its screen output.
- **One look at each test set.** Each model saw each test set once; the second improved model's test results were read only after the validation decision.

12. Discussion

Data mattered more than resolution, again. Milestone 2 found that label definitions mattered more than the architecture. Milestone 3 found the same pattern in a new form: the large gains came from new kinds of training data and complete labels, while a 56% increase in input pixels gave nothing measurable. A model learns the conditions it is shown, and it can only be as precise as the boxes it is shown.

The diagnosis earned its cost. Measuring on validation first turned a vague goal ("more accurate") into specific targets (rotation, backgrounds, partial boxes) and argued against an expensive one (resolution for small objects). The weakness it found most clearly, rotation, is the one the milestone fixed most completely.

Honest selection can disagree with intuition. Ultralytics' validation suggested the newer, larger model was better; the rule, applied to the same validation set with the same code, said the opposite with a clear

margin. Fixing the rule in advance made it possible to accept that answer without the suspicion that it had been chosen.

12.1 Threats to validity

- **Synthetic scenes.** Scattered and cut-out test scenes use unseen photos and unseen backgrounds but are built like the training scenes, so their scores are somewhat optimistic for real photographs of several objects.
- **A test-informed idea.** The idea of class-agnostic duplicate removal came from a rise in wrong IDs on the first improved model's cut-out test (57 to 66); the setting for the final model was chosen on validation only, but the motivation was seen on test.
- **Corrected truth.** The test truth was corrected in this milestone (17 test boxes in pass 2 alone), so results are not directly comparable with the Milestone 2 report; all comparisons here use the same corrected truth.
- **Confounded comparison.** boost and boost2 differ in labels, input size and batch size at once, so the result does not isolate the effect of resolution.
- **One run per configuration.** No model was trained with several seeds, so differences of a few thousandths between configurations are within run-to-run variation; the selection margin is larger than the bootstrap's image-level uncertainty, but seed-level variation was not measured.
- **Validation used twice.** Each model's best epoch was chosen on validation and the selection was made on validation, which makes validation scores slightly optimistic for both models alike.
- **The explorer's live grids.** The EYE scores random grids, and therefore the stress tests of Section 8.5, against the Milestone 2 box file; the label corrections of this milestone are used in all evaluation numbers but not yet in the explorer's live scoring, which can therefore differ on a few test photos.

13. Limitations and Future Work

Two measurements would complete the picture. First, a stress test of both models on degraded images: Milestone 2 traded some robustness to heavy degradation for accuracy on normal images (mAP@0.5 0.506 to 0.428), and this milestone added low light, blur, JPEG and noise to every kind of training image, so the gain there is likely but unmeasured. Second, an evaluation on real photographs of several course objects together, which would show how far the synthetic scenes transfer. Look-alikes under rotation remain the main source of wrong IDs; more training photos of these groups or a larger model (YOLO26-m, about twice the training time) are the likeliest remedies, and an ensemble of the two improved models would probably add a little at the cost of speed. Oriented boxes would remove the loose boxes of tilted objects, and refining boxes with SAM masks would raise the precision limit of the labels, but both would change the answer key and require new verification. For The EYE, the explorer's live scoring should use the corrected box file, and a permanent host would remove the dependence on the laptop.

14. Conclusion

Milestone 3 took a detector that worked on upright grids and made it trustworthy beyond them. A person checked every usable box in the dataset; new training data taught rotation, scattered layouts and arbitrary backgrounds; a rule fixed in advance chose between two improved models on validation; and validated settings for grids and scenes completed the system. On five test sets with corrected truth, the selected model

finds 95% of objects on normal grids and single photos, 92% on rotated grids, 90% on scattered scenes and 88% on objects placed on unseen backgrounds, against 45% to 91% for the Milestone 2 model, with the remaining weaknesses stated: tiny and dark objects, look-alikes under rotation and a precision limit set by the labels. The EYE now shows all of this on every layout and model, explains each result, and can be opened by anyone in the course through a link.

Code: The GitHub repository (github.com/tanmayi123/Discriminative-Project-Deep-Learning) contains code only. Images, grids, labels and model weights are kept out of the repository, as required.

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References 1 to 17 are those of the Milestone 2 report; this report adds 18 to 22.

Appendix A. Files and Reproducibility

Hardware and software are unchanged from Milestone 2 (NVIDIA RTX 5070 Laptop GPU, 8 GB; PyTorch 2.11; Ultralytics 8.4). All new code and outputs are in `Milestone 2\M2 extra`; the hosting files are in `Milestone 3\M3 HOST`. As in Milestone 2, the repository holds code only.

Table 12. Scripts of Milestone 3.

File	Role
`m2_diagnose.py`	validation diagnosis: rotation, miss causes, levers without retraining
`m2_label_check.py`	the review tool: prepare, review, apply, merge; second pass with the improved model
`m2_boost.py`	SAM 2 masks, training data and test sets, training (resumable), evaluation of all models on the same truth
`m2_boost_val.py`	model selection by the fixed rule with a paired bootstrap; grid and scene settings on validation; one look at the tests
`m2_eye_app_v3.py`	The EYE v3 (port 7862); v2 (7861) and the original (7860) unchanged
`start_share.ps1`, `stop_share.ps1`	sharing through a Cloudflare quick tunnel
`make_space_bundle.py`, `upload_space.py`	self-contained bundle and upload for a Docker Space

Table 13. Recorded hashes (SHA-256, first and last characters).

File	Purpose	Hash
`selection_rule.txt`	model-selection rule, fixed before any boost2 result	C6A67BD9…F8C0
`m2_boost_val.py`	selection and post-processing script, fixed before any boost2 result	6250EFDE…5442
Milestone 2 `best.pt`	protected fallback model	90C434C2…BB41
Milestone 2 `results.json`	protected results	800A8F0E…F007
Milestone 2 `boxes_master.csv`	protected box file	D8B94B64…8F96
Milestone 2 `m2_eye_app.py`	protected explorer	863FB939…F072

Appendix B. Complete Evaluation Tables

Table 14. The Milestone 2 model and boost2 (bold), same corrected truth (from boost2's evaluation). mAP50 = mAP@0.5, mAP50-95 = mAP@0.5:0.95.

Test set	Model	Recall	Precision	Wrong IDs	Fully correct	Small-object recall	mAP50	mAP50-95
Normal grids	M2 model	0.912	0.950	16	0.347	0.750	0.947	0.873
	boost2	**0.947**	**0.976**	**9**	**0.458**	**0.844**	**0.972**	**0.897**
Rotated grids	M2 model	0.756	0.845	230	0.093	0.458	0.813	0.714
	boost2	**0.912**	**0.963**	**31**	**0.306**	**0.712**	**0.953**	**0.858**
Scattered	M2 model	0.767	0.830	33	0.086	0.411	0.803	0.635
	boost2	**0.896**	**0.936**	**14**	**0.279**	**0.681**	**0.942**	**0.794**
Cut-outs	M2 model	0.452	0.799	70	0.000	0.157	0.559	0.527
	boost2	**0.871**	**0.896**	**53**	**0.183**	**0.602**	**0.904**	**0.873**
Single photos	M2 model	0.905	0.954	12	0.904	0.698	0.941	0.865
	boost2	**0.940**	**0.977**	**6**	**0.940**	**0.844**	**0.966**	**0.891**

Table 15. boost and boost2 (bold), each at its single validated threshold, same tests. mAP50 = mAP@0.5, mAP50-95 = mAP@0.5:0.95.

Test set	Model	Recall	Precision	Wrong IDs	Fully correct	Small-object recall	mAP50	mAP50-95
Normal grids	boost	0.942	0.977	7	0.444	0.875	0.971	0.900
	boost2	**0.947**	**0.976**	**9**	**0.458**	**0.844**	**0.972**	**0.897**
Rotated grids	boost	0.909	0.964	38	0.315	0.701	0.953	0.863
	boost2	**0.912**	**0.963**	**31**	**0.306**	**0.712**	**0.953**	**0.858**
Scattered	boost	0.889	0.944	14	0.250	0.652	0.936	0.803
	boost2	**0.896**	**0.936**	**14**	**0.279**	**0.681**	**0.942**	**0.794**
Cut-outs	boost	0.876	0.898	54	0.240	0.566	0.912	0.887
	boost2	**0.871**	**0.896**	**53**	**0.183**	**0.602**	**0.904**	**0.873**
Single photos	boost	0.940	0.976	8	0.939	0.865	0.970	0.894
	boost2	**0.940**	**0.977**	**6**	**0.940**	**0.844**	**0.966**	**0.891**

Table 16. Selected model with the final settings on every test set (thresholds 0.10 on grids and single photos, 0.35 with duplicate removal on scenes).

Test set	Threshold	Images	Objects	Recall	Precision	Wrong IDs	Fully correct	Small-object recall
Normal grids	0.10	72	1,048	0.949	0.970	10	0.472	0.896
Rotated grids	0.10	216	3,144	0.922	0.947	58	0.347	0.747
Single photos	0.10	1,040	1,048	0.951	0.968	11	0.951	0.896
Scattered	0.35	104	1,039	0.897	0.943	17	0.250	0.667
Cut-outs	0.35	104	1,014	0.881	0.914	61	0.288	0.590

Appendix C. The EYE User Manual

This appendix is the manual written for teammates and visitors. It documents every control, colour and number on the screen of The EYE v3; each behaviour was checked against the code.

C.1 Start and stop

1. Open a terminal in the project folder `PROJECT - 1\Milestone 2\M2 extra` and close anything else using the GPU, such as a training run or a second EYE.
2. Start the server with `python m2_eye_app_v3.py`. The browser opens http://127.0.0.1:7862; if it does not, type the address.
3. Wait for the intro animation and for the status line under the eye to say *model ready*. The first detection also loads the model and takes a few seconds; later ones take well under a second.
4. To stop, press Ctrl+C in the terminal, or close it.

Table 17. Versions of The EYE.

Version	Start with	Address	Use it for
EYE v3	`python m2_eye_app_v3.py` in `Milestone 2\M2 extra`	:7862	everything; the final version
EYE v2	`python m2_eye_app_v2.py` in `Milestone 2\M2 extra`	:7861	fallback with the first improved model
Original	`python m2_eye_app.py` in the Milestone 2 folder	:7860	fallback with the Milestone 2 model only

C.2 The screen

Table 18. Areas of the screen (Figure 11).

Area	Holds
Brow (top)	six set-up panels: Grid, Mix, Conditions, Seed, Model, Threshold
Left dock	Upload, Random, Stress, About
Pupil	the test image with its boxes; click it for the large view
Iris ring	the DETECT button, then one coloured segment per object
Right dock	Files, Snapshot, Titles (hover hints), Light or Dark theme, Calm (no motion)
Review panels (bottom)	Score, Results, Inspect, and Threshold sweep or Box tightness
Top corners	"How was I built?" (left); dataset counts (right), with the reasons for left-out photos on hover

C.3 Making a test image

Random images use only held-out test photos whose boxes were checked by a person, so every box can be marked right or wrong. An image can also be uploaded, or dragged anywhere onto the page.

Table 19. Set-up choices on the brow.

Panel	Choices	Effect
Grid	1x1, 2x2, 3x3, 4x4, 5x4, 5x5	1 to 25 photos side by side (224 to 1,120 px per side)
Mix: sections	one chip per section of the object list	only objects of the chosen sections go into grids
Mix: selection	random, look-alikes	random objects, or whole look-alike groups together (watches, wallets, rackets, power banks, sunglasses and goggles and more)
Mix: layout	grid, rotated, scattered, cut-outs	an upright grid; the same grid turned 90°, 180° or 270°; or one of the evaluation's prepared scenes (104 each) on unseen backgrounds
Conditions	none, blur, low-res, dark, jpeg, all	damage the image before detection; "all" is the low-quality test of the Milestone 2 report
Seed	empty, a number, random, keep last	the same seed and settings give the same image on every laptop; "random" clears the box, "keep last" copies the seed on screen

Scenes are prepared, not built: grid size, sections and selection do not apply to them, and the seed picks which scene appears. Uploads cannot be scored, because there are no true boxes for them; an upload whose sides are multiples of 224 px is treated as a grid, any other as a scene, and photos larger than 3,000 px are scaled down first.

C.4 Model, threshold and detection

- **Model.** Every model whose weights are on the laptop; "selected" marks the model chosen on validation (yolo26s_boost), "M2 pick" the Milestone 2 model (yolo26s_ft2).
- **Threshold.** A slider from 0.05 to 0.90. Each model has validated values, for the selected model 0.10 on grids and 0.35 on scenes; while the slider is untouched, The EYE uses the right one for the image on screen and says so in the note under the button. A value set by hand is kept until *reset to validated*.
- **One box per cell.** On grids only the most confident box in each 224-px cell is kept, which removes most extra boxes. Scenes use the selected model's rule instead: where boxes of any name overlap heavily, only the most confident is kept.
- **Detect.** Press the DETECT plate on the iris ring or the D key.

C.5 Reading the result

Table 20. The five results and how they are drawn.

Result	Drawn as	Meaning
correct	solid green box	right object, and the box overlaps the true box by at least half (IoU ≥ 0.50)
box off	dashed amber box	right object, but an overlap below 0.50
wrong ID	solid magenta box	the box is on the object but names another one; look-alikes are flagged
missed	dashed cyan box at the true position	no box for a real object; the reason says whether it was seen below the threshold, seen weakly as something else, or very small
extra	dotted grey box	a box that matches no object
true box	thin dashed white	where the object really is, as checked by a person

Each box carries the Object ID and its confidence, for example OBJ042 0.97. Hovering shows the name, section, class AP, look-alike group and the reason for an error; clicking selects the object everywhere. After detection the iris ring shows one segment per object in these colours, starting at the left of the DETECT plate and running clockwise.

C.6 The review panels

Table 21. The four review panels.

Panel	Shows	How to use it
Score	objects correct out of all; FULLY CORRECT or k/n NAMED CORRECTLY; counts per result and per section; time	the one-line verdict for the image
Results	one sentence per object, filtered to errors or all	click a line to zoom the pupil onto it; + opens the full record (result, reason, true and predicted object, section, look-alike group, class AP, confidence, IoU, both boxes)
Inspect	the object cut from the image beside training photos of the predicted and the true class	see why a look-alike was confused; step with ‹ ›
Threshold sweep	correct, missed, extra and wrong ID at every threshold from 0.05 to 0.90 (Figure 14, panel b)	if misses fall just left of the current threshold, a slightly lower one would find them
Box tightness	one dot per boxed object: IoU across, confidence up, with the 0.50, 0.75 and 0.95 cut-offs (panel a)	mAP@0.5:0.95 on one image: dots left of a line lose credit there

C.7 Large view, stress test and files

- **Large view.** Click the pupil: grid, seed, model and threshold can be changed at the top, Random and Detect run in place, and the side shows the legend, a summary, the object list and the downloads (Figure 12).
- **Stress test.** Runs 10 to 300 random grids (100 by default; capped when the explorer is shared) with the brow settings, from seeds 1000, 1001, ..., so a run repeats exactly. It reports objects correct, fully

correct grids with a 95% interval, named correctly, time per grid, the error counts, the objects most often missed or wrong, the confusions and the failure reasons. It always builds grids.

- **Files.** After a detection: the annotated image (PNG), and one row per object (CSV and JSON) with result, reason, objects, section, class AP, confidence and IoU; for uploads, one row per box with its corners. **Snapshot** makes a report-ready picture with the image, the score and every error in words.
- **About and "How was I built?".** About holds every number, including the before-and-after tables on every test set and the selection result. "How was I built?" tells the project in twelve flip cards; arrow keys move between stations and Enter flips a card.

C.8 Keyboard and mouse

Table 22. Shortcuts.

Key or action	Effect
N	new random image
D	detect
U	upload a photo
Esc	close a window, or zoom out and clear the selection
← →, Enter	move between journey stations, flip a card
click the pupil	large view
drag a photo in	upload it
⤢ on a panel	enlarge it

C.9 Troubleshooting

Table 23. Common situations.

You see	Do this
the page does not open	check the terminal; if the port is busy, an older copy is running: close it and start again
the first detection is slow	normal: the model loads then
every detection is slow	something else is using the GPU, often a second EYE or a training run
an upload shows no score	expected: uploads have no true boxes
nothing detected on an upload	try a lower threshold, or a photo in which the object is larger and upright
look-alikes refuses a large grid	choose a smaller grid or more sections
hard to read on a projector	zoom the browser to 125 to 150%, try the light theme, switch on Calm